

Effect of large bore mechanical thrombectomy on pulmonary vascular resistance in patients with acute pulmonary embolism

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Aims

In patients with intermediate-risk pulmonary embolism (PE), there are limited tools to assess therapeutic response following catheter-based intervention. This study evaluates pulmonary vascular resistance (PVR), an invasive marker of right ventricular (RV) afterload, and its prognostic significance in acute PE.

Methods and results

This single-centre retrospective study included patients from October 2020–May 2025 with intermediate-high risk PE undergoing large bore mechanical thrombectomy (LBMT) with pulmonary artery catheter-derived haemodynamic indices obtained pre- and post-procedure. The primary objective was to evaluate the effect of LBMT on PVR. The secondary objective was to evaluate the predictors of post-procedure elevated PVR (defined as PVR >2 Wood units, WU) and its effect on clinical composite outcome (PE mortality, resuscitated cardiac arrest, haemodynamic instability, and 90-day hospital readmission) and hospital length of stay (LOS). A total of 131 patients were included. Following LBMT, median PVR decreased significantly from 2.9 to 1.8 WU ($P < 0.001$), with greater reduction in patients with higher baseline PVR (baseline PVR tertile 3 to 1: 50% vs. 40% vs. 20%; $P < 0.001$). Persistently elevated post-procedure PVR (>2 WU) was seen in 44% of patients. However, the incidence of post-procedure severe PVR >5 WU was extremely low (11.5% pre-procedure, 0.8% post-procedure). Multivariable predictors of elevated post-procedural PVR were pre-procedural mean pulmonary artery pressure (OR: 1.07, 95% CI 1.01–1.14, $P = 0.026$) and pre-procedural PVR (OR 2.20, 95% CI: 1.20–4.04, $P = 0.011$). In an age and sex adjusted model, elevated post-procedure PVR was associated with a longer in-hospital LOS of 4.2 days (95% CI: 0.60–7.88; $P = 0.023$) and a 4-fold higher risk of the composite outcome (20.7% vs. 5.3%, adjusted hazard ratio: 4.02, 95% CI: 1.28–12.61, $P = 0.017$).

Conclusion

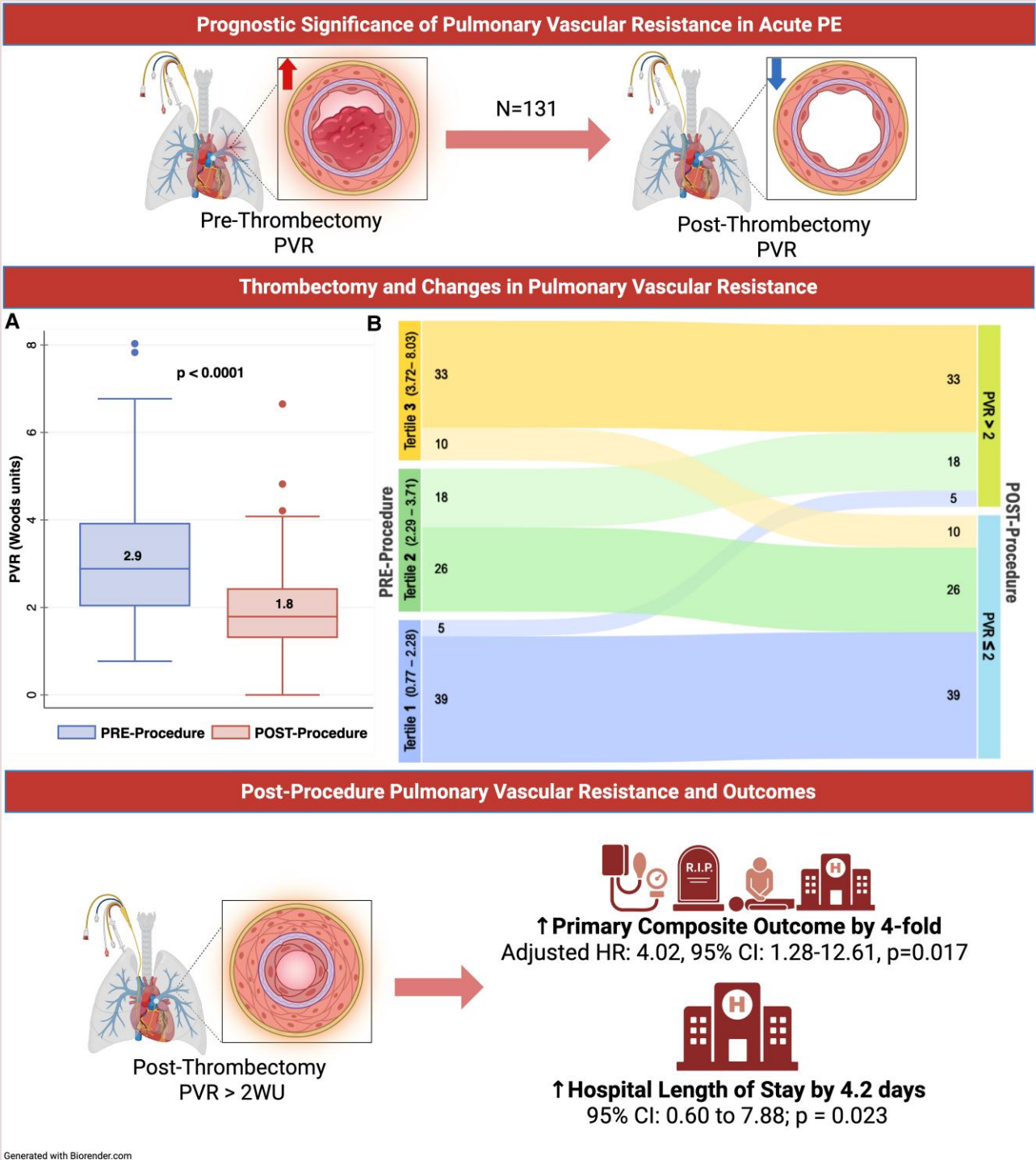
In patients with intermediate-high risk PE, LBMT significantly reduced PVR and may be a valuable haemodynamic marker of disease severity and treatment response. Elevated post-procedural PVR identified patients at increased risk of adverse outcomes.

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Graphical Abstract



Keywords Pulmonary embolism • Right ventricular failure • Mechanical thrombectomy • Haemodynamics

Introduction

Patients with intermediate-risk pulmonary embolism (PE), defined by right ventricular (RV) dysfunction on imaging and/or elevated biomarkers in the absence of overt circulatory failure, experience widely variable outcomes, with reported mortality rates ranging from 2% to 15%, and a notable proportion experiencing clinical deterioration.^{1,2} With an emerging, albeit presently limited robust evidence base, a significant percentage of these patients undergo catheter-based therapies (CBT).^{3,4} However, identifying those most likely to benefit from intervention and defining procedural endpoints remains a major clinical unmet need.

Contemporary studies of intermediate-risk PE patients treated with CBT report very low mortality and complication rates.^{5–7} Yet, these outcomes likely reflect altered disease trajectories and selection bias. In this context, surrogate endpoints, particularly invasive haemodynamic parameters, may better reflect therapeutic response. Direct obstruction of the pulmonary arteries, hypoxaemia, and potent pulmonary arterial vasoconstrictors cause a rapid increase in pulmonary vascular resistance (PVR) and thus RV afterload. Sudden elevation in RV afterload may result in RV hypokinesis and dilation, tricuspid regurgitation, and, ultimately, RV failure and shock spiral. This makes PVR a central determinant of haemodynamic consequences of acute PE.⁸

However, the extent of PVR reduction achieved with LBMT and its clinical implications have not been studied in detail. To address this gap, we evaluated changes in PVR following LBMT in patients with intermediate-risk PE. In addition, we examined predictors of persistently elevated post-procedural PVR and its association with clinical outcomes.

Methods

Study design and patient population

Consecutive adult patients (≥ 18 years) with intermediate-high risk pulmonary embolism (PE) who underwent LBMT at a large quaternary academic medical centre between October 2020 and May 2025 were included in this retrospective analysis.⁹ All patients had symptomatic, proximally located PE confirmed by computed tomography (CT), and the decision to pursue intervention was made following multidisciplinary review by the Pulmonary Embolism Response Team (PERT). Intermediate-risk classification was based on criteria outlined by the European Society of Cardiology (ESC) guidelines.⁸ This study was approved by the New York University Institutional Review Board.

Measures and outcomes

The primary haemodynamic outcome was changes in PVR post procedure. Our secondary objective was to identify predictors of high-post procedure PVR (defined as $\text{PVR} > 2$ WU) as well as to evaluate whether a residual high PVR was a predictor of clinical outcomes in this cohort. At our institution, patients treated with LBMT routinely undergo pulmonary artery catheterization with invasive haemodynamic variables including pulmonary artery pressures and cardiac index (CI) recorded immediately prior to, and post-intervention, in accordance with standard of care practices. PA pressures were measured consistently in the main pulmonary artery pre and post-procedure. The CI was estimated according to the indirect Fick method, with a CI of ≤ 2.2 L/min/m² used as a cutoff to define a low value.¹⁰ PVR was calculated as [mean pulmonary artery pressure—pulmonary artery diastolic pressure (PADP)]/cardiac output. Due to the lack of measurement of pulmonary capillary wedge pressure (PCWP) in our cohort, PADP was used as a surrogate and has been shown to have good correlation with PCWP.¹¹ Furthermore,

acute PE patients undergoing LBMT typically have a significant thrombus burden, which presents logistical challenges of obtaining a reliable PCWP. Pulmonary compliance was calculated as Stroke Volume/Pulmonary Artery Pulse Pressure.¹² PA compliance represents the distensibility of the pulmonary vasculature and serves as a complementary marker to PVR. Total pulmonary arterial elastance (E_a), another index of right ventricular afterload that integrates both resistive and pulsatile properties, was calculated as $(\text{PASP} \times 0.9)/(\text{cardiac output/heart rate})$.¹³ All procedural data were timestamped and recorded in an electronic patient log.

All CT and echocardiographic variables were independently measured and calculated by 2 investigators blinded to the clinical data. Baseline demographics, echocardiographic data, invasive haemodynamics, and hospital clinical outcomes were collected and analysed. RV dysfunction was defined by one or more of the following: RV dilation or hypokinesis on echocardiogram, or right-to-left ventricular internal diameter ratio (RV:LV) > 0.9 using the internal diameter/minor axis of RV and LV in the transverse plane.¹⁴ The Simplified Pulmonary Embolism Severity Index (sPESI) and Composite Pulmonary Embolism Shock (CPES) scores were calculated using methods from the original derivation and validation studies.^{15–18} Echocardiographic parameters were defined according to current guidelines and prior studies.^{19,20}

The primary clinical outcome was a composite of 30-day PE-related mortality, resuscitated cardiac arrest, haemodynamic instability, and 90-day readmission rates. Outcomes were adjudicated by two physicians blinded to tertiles of PVR.

Statistical analysis

Continuous variables were presented as mean $\pm$ standard deviation (SD) or median with interquartile range (IQR) for skewed data. Categorical data were expressed as frequencies and proportions. Student's *t*-test was utilized to compare paired continuous variables when assumptions were met for normal distribution, and the Wilcoxon signed-rank test was used when the distribution was not normal. Categorical variables were compared using χ^2 test or the Fisher exact test. Patients were stratified into tertiles based on pre-procedural PVR: Tertile 1 (T1): 0.77–2.28 Wood units (WU), Tertile 2 (T2): 2.29–3.71 WU, and Tertile 3 (T3): 3.72–8.03 WU. Regression models were used to identify predictors of post-procedural elevated PVR (> 2 WU) and predictors of hospital and ICU length of stay. To assess multi-collinearity in the multivariable model, we calculated the variance inflation factor, with a variance inflation factor value greater than 10 considered indicative of significant multicollinearity. We employed a Cox proportional hazard model to assess the relationship between the clinical outcomes and PVR, adjusting for age and sex. Kaplan–Meier curve was generated for the two groups based on the post-procedural residual elevated PVR (> 2 vs. ≤ 2 WU). The log-rank test was used to compare outcomes between the two groups. All tests were considered significant at a two-sided alpha level < 0.05 . All analyses were performed using Stata 18 software (StataCorp LP, College Station, TX).

Results

Baseline patient characteristics

A total of 131 patients with intermediate-high risk PE who underwent LBMT were stratified into tertiles based on pre-procedural PVR. Patients in the highest PVR tertile (T3, $n = 43$) exhibited several distinct clinical, biomarker, imaging, and haemodynamic features compared to those in the lower tertiles (T1 and T2, $n = 44$ each). Patients in tertile 3 had significantly lower oxygen saturation on presentation, higher levels of lactate, troponin, and BNP, consistent with greater myocardial injury and RV strain (Table 1). Imaging markers of right heart strain increased across tertiles. Median RV/LV diameter ratio on CT was 1.7 in tertile 3, 1.4 in tertile 2, and 1.1 in tertile 1

($P < 0.001$). Patients in tertile 3 had higher PASP on echocardiography and lower TAPSE when compared with the other tertiles. Similarly, pre-procedural invasive mPAP and PASP were highest in tertile 3 with lowest CI (Table 2) when compared with the other tertiles. All patients in the highest PVR tertile had low cardiac index (≤ 2.2 L/min/m²), compared to 61% in the middle tertile and 23% in the lowest tertile ($P < 0.001$).

Haemodynamic changes

Following LBMT, there was a significant reduction in mPAP across the three tertiles (T3 to T1: -8.0 , -6.0 , -4.0 mmHg; $P = 0.015$), with patients in the highest PVR tertile experiencing the greatest improvement (Supplemental material online, Figure S1). Similarly, there was a significant reduction in PASP across the 3 tertiles (T3 to T1: -11.0 , -14.5 , -7.5 mmHg; $P < 0.001$). Moreover, CI improved significantly across the 3 tertiles (Table 2).

Pulmonary vascular resistance

The majority of patients (75.2%) had an elevated PVR (i.e. >2.0 Wood Units) with 15 (11.5%) having severely elevated PVR (>5 WU) at baseline. The median PVR decreased from 2.9 to 1.8 WU [38% relative reduction (RR)] reflecting marked improvement in right ventricular afterload following intervention (Figure 1). There was a greater PVR reduction in patients with higher baseline PVR (baseline PVR tertile 3 to 1: 50% vs. 40% vs. 20% RR; $P < 0.001$). Persistently elevated post-procedure PVR (>2 WU) was seen in 43.6% of patients. However, the incidence of post-procedural severe residual PVR (>5 WU) was exceedingly uncommon (0.8%, $n = 1$). A greater proportion of patients in tertile 3 had an elevated post-procedural PVR >2 (77%) compared to tertile 2 (41%) and tertile 1 (11%; $P < 0.001$).

Pulmonary artery compliance

The majority of patients (76.3%) had low PA compliance (defined as <2 mL/mmHg) pre-procedure. Pre-procedure PA compliance was lower with higher PVR tertiles (Table 2). PA compliance improved significantly following LBMT, rising from a median value of 1.3 to 2.3 mL/mmHg ($P < 0.001$, 77% increase). This improvement was consistent across PVR tertiles (T3: 90%, T2: 80%, T1: 40%) and paralleled the reduction in PVR, though the relative magnitude of change in compliance was substantially greater than that of PVR. Post LBMT, 41 patients (31.3%) had residual low PA compliance.

Among patients with elevated post-procedure PVR (>2 WU), post-procedure PA compliance was frequently normal. As illustrated in Figure 2, 24 patients (43%) with PVR >2 WU had a preserved PA compliance (≥ 2 mL/mmHg).

Pulmonary artery elastance

Median pre-procedure E_a progressively increased across PVR tertiles (Table 2, $P < 0.001$). Similarly, absolute decreases and percentage change in PA elastance were more pronounced in the higher tertiles ($P < 0.001$). An elevated pre-procedure E_a (> 0.5 mm Hg/mL) was seen in 91.6% of the cohort vs. 68% of the cohort post-procedure ($P < 0.001$).

Outcomes

ICU and hospital LOS were similar across tertiles (Table 2). There were no differences with regard to the primary clinical composite outcome or the individual components of the primary outcome between the tertiles.

The relationship between post-procedural PVR and the primary composite outcome is illustrated in a restricted cubic spline curve in Supplementary material online, Figure S2. Post-procedural PVR >2 WU was associated with a longer in-hospital LOS of 4.2 days (95% CI: 0.60–7.88; $P = 0.023$) after adjusting for age and sex. After a mean follow-up time of 72 days, 16/131 (12%) patients experienced the primary composite outcome. In a multivariable model after adjusting for age and sex, post-procedural PVR was associated with a higher risk of the primary composite outcome (adjusted HR: 1.58, 95% CI: 1.11–2.24, $P = 0.01$), driven largely by 90-day readmission with very low risk of PE-related mortality. Notably, a post-procedural PVR >2 WU was associated with a 4-fold increased risk of the primary composite outcome [20.7% vs. 5.3%, aHR: 4.02, 95% CI: 1.28–12.61, $P = 0.017$, (Figure 3)]. Per cent PVR change was also associated with the primary outcome (adjusted HR 0.83, 95% CI: 0.70–0.98, $P = 0.035$, every 10% change), while pre-procedure PVR was not associated with the primary outcome (adjusted HR: 1.29, 95% CI: 0.96–1.73, $P = 0.088$). Of note, even in PVR tertile 3, where 77% of patients had post-procedure residual PVR >2 WU, the in-hospital mortality was 0%.

Predictors of elevated post-procedural PVR

In univariable logistic regression, several clinical and haemodynamic parameters were significantly associated with elevated post-procedural PVR (Table 3). After adjusting for covariates, only pre-procedural mPAP (OR: 1.07; 95% CI: 1.01–1.14; $P = 0.026$) and pre-procedural PVR (OR: 2.20; 95% CI: 1.20–4.04; $P = 0.011$) remained independently associated with persistently elevated post-procedural PVR.

Discussion

To our knowledge, this is the first study to evaluate the effect of LBMT on PVR in patients with acute intermediate-high risk PE. LBMT significantly reduced PVR across tertiles of baseline PVR, with a greater reduction in higher tertiles. Persistently elevated post-procedure PVR >2 WU was seen in 44% of patients. Among these patients, 43% had normal PA compliance, reflecting faster improvement in PA compliance when compared with improvements in PVR. Elevated post-procedural PVR was associated with longer hospital LOS and a higher risk of the primary composite outcome. Additionally, every 10% decrease in PVR was associated with a 17% lower risk in the primary outcome at follow-up. Finally, severe residual PVR post-thrombectomy was exceedingly uncommon.

There has been increasing interest in CBT in patients with acute PE due to its rapid ability to remove clot burden and unload the RV, improve haemodynamics, and prevent progressive RV failure, the central cause of early mortality in acute PE.^{2,21} While patient selection for CBT remains an unresolved issue, an additional challenge is the lack of clearly defined procedural endpoints for LBMT.²² Unlike coronary or peripheral arterial interventions, where angiographic or flow-based endpoints are well established,²³ no consensus exists for when to conclude a pulmonary thrombectomy procedure. Operators currently rely on surrogate markers such as improvement in haemodynamics, including blood pressure and heart rate, and angiographic assessments of residual clot burden. Moreover, the degree of haemodynamic improvement can be heterogeneous and may not correlate with the amount of clot removed. Residual pulmonary vascular obstruction is a recognized risk factor for adverse outcomes, including in chronic thromboembolic

Table 1 Clinical characteristics stratified across baseline PVR tertiles

	Tertile 1 (n = 44)	Tertile 2 (n = 44)	Tertile 3 (n = 43)	P-value
Demographics				
Age	54.0 (38.5, 65.5)	61.0 (55.0, 69.5)	58.0 (49.0, 72.0)	0.098
Race				0.65
White	5 (11%)	2 (5%)	6 (14%)	
Black/African American	29 (66%)	29 (67%)	26 (60%)	
Hispanic or Latino	5 (11%)	7 (16%)	6 (14%)	
Asian	1 (2%)	1 (2%)	1 (2%)	
Women	25 (57%)	20 (45%)	18 (42%)	0.34
Hypertension	18 (41%)	27 (61%)	15 (35%)	0.034
Diabetes	8 (18%)	16 (36%)	6 (14%)	0.030
CKD (III-V)	4 (9%)	8 (18%)	1 (2%)	0.046
Chronic Lung Disease	5 (11%)	5 (11%)	7 (16%)	0.73
BMI (kg/m ²)	32.8 ± 8.6	33.7 ± 7.5	31.2 ± 6.9	0.33
Morbid Obesity (BMI >40 kg/m ²)	9 (20%)	6 (14%)	4 (9%)	0.33
Malignancy	5 (11%)	11 (25%)	4 (9%)	0.085
Clinical Presentation				
Heart Rate (bpm)	110.0 (96.5, 118.0)	105.5 (90.0, 119.5)	116.0 (105.0, 123.0)	0.12
Systolic BP (mm Hg)	127.5 (116.5, 144.0)	126.0 (110.0, 141.0)	119.0 (107.0, 138.0)	0.35
Diastolic BP (mm Hg)	84.5 (70.0, 95.5)	79.0 (68.5, 91.0)	78.0 (70.0, 90.0)	0.47
SpO ₂ (%)	97.0 (92.5, 99.0)	93.0 (89.0, 96.0)	93.0 (91.0, 98.0)	0.011
Syncope	9 (21%)	8 (18%)	11 (26%)	0.70
WBC (cells/μL)	8.9 (7.0, 11.1)	9.0 (7.4, 12.0)	10.4 (8.4, 13.2)	0.062
Hemoglobin (g/dL)	12.1 (11.0, 13.6)	13.2 (11.7, 13.9)	13.6 (10.4, 14.9)	0.13
Creatinine (mg/dL)	0.9 (0.8, 1.2)	1.0 (0.9, 1.3)	1.0 (0.8, 1.3)	0.35
Lactate (mmol/L)	1.9 (1.2, 2.8)	2.6 (1.7, 3.5)	2.2 (1.7, 3.9)	0.041
HS-Troponin (ng/L)	79 (33, 153)	145.5 (67, 249)	155.5 (94, 393)	0.008
BNP (pg/mL)	934 (397, 1673)	1850 (561, 4400)	2000 (523, 5948)	0.031
Proximal DVT	27 (61%)	30 (68%)	29 (67%)	0.32
Pulmonary Embolism (PE) Characteristics				
CPES Score	5.0 (4.0, 5.5)	5.0 (4.0, 5.0)	5.0 (4.0, 6.0)	0.38
sPESI	1.0 (0.0, 1.0)	1.0 (0.5, 2.0)	1.0 (1.0, 2.0)	0.26
Echocardiography				
LVEF (%)	65.0 (57.0, 75.0)	65.0 (55.0, 75.0)	70.0 (60.0, 75.0)	0.59
TAPSE (cm)	1.8 (1.4, 2.2)	1.6 (1.1, 2.1)	1.3 (1.1, 1.6)	0.006
PASP (mmHg)	37.0 (33.0, 44.0)	51.5 (34.0, 58.0)	54.7 (43.0, 64.0)	0.001
McConnell Sign	25 (58%)	29 (72%)	26 (67%)	0.14
Computed Tomography (CT)				
Saddle PE	20 (45%)	17 (41%)	23 (53%)	0.53
PE Location				0.68
Main PA	40 (95%)	38 (93%)	37 (90%)	
Lobar	2 (5%)	3 (7%)	4 (10%)	
RV/LV Diameter Ratio	1.1 (1.0, 1.5)	1.4 (1.2, 1.8)	1.7 (1.4, 2.0)	< 0.001

All values are presented as frequency (%) or median (IQR). Bolded values denote statistically significant p-values.

Tertile 1: 0.77–2.28 Wood units (WU), Tertile 2: 2.29–3.71 WU, Tertile 3: 3.72–8.03 WU.

BMI: Body Mass Index; BNP: B-type Natriuretic Peptide; BP: Blood Pressure; CKD: Chronic Kidney Disease; CPES: Composite Pulmonary Embolism Severity; DVT: Deep Vein Thrombosis; HS: high-sensitivity; LV: Left Ventricle; LVEF: Left Ventricular Ejection Fraction; PA: Pulmonary Artery; PASP: Pulmonary Artery Systolic Pressure; PVR: Pulmonary Vascular Resistance; RV: Right Ventricle; sPESI: Simplified Pulmonary Embolism Severity Index; TAPSE: Tricuspid Annular Plane Systolic Excursion; WBC: White Blood Cell count. Tertile 1: 0.77–2.28 WU, tertile 2: 2.29–3.71 WU, and tertile 3: 3.72–8.03 WU.

Table 2 Invasive hemodynamics and clinical outcomes

	PVR Tertile 1 (n = 44)	PVR Tertile 2 (n = 44)	PVR Tertile 3 (n = 43)	P-value
Pre-Thrombectomy				
mPAP (mmHg)	28.5 (22.0, 33.5)	32.0 (26.5, 39.0)	36.0 (29.0, 45.0)	<0.001
PASP (mmHg)	45.0 (38.5, 52.0)	54.0 (49.0, 63.5)	61.0 (52.0, 73.0)	<0.001
SvO ₂ (%)	69.0 (63.1, 74.0)	61.5 (54.8, 66.5)	52.0 (48.0, 61.0)	<0.001
CI (l/min/m ²)	2.6 (2.3, 3.0)	2.1 (1.9, 2.4)	1.7 (1.4, 1.9)	<0.001
CI ≤2.2 L/min/m ²	10 (23%)	27 (61%)	43 (100%)	<0.001
PVR	1.9 (1.5, 2.0)	2.9 (2.5, 3.3)	4.3 (4.0, 5.4)	<0.001
PA Compliance (mL/mmHg)	2.2 (1.9, 2.9)	1.3 (1.1, 1.7)	0.9 (0.8, 1.0)	<0.001
PA Compliance (<2 mL/mmHg)	15 (34%)	42 (95%)	43 (100%)	<0.001
PA Elastance	0.6 (0.5, 0.8)	1.1 (0.8, 1.3)	1.6 (1.4, 2.0)	<0.001
Post-Thrombectomy				
mPAP (mmHg)	23.0 (19.0, 29.5)	26.0 (21.5, 31.0)	28.0 (23.0, 35.0)	0.027
Δ mPAP	−4.0 (−7.5, 0.0)	−6.0 (−9.0, −3.0)	−8.0 (−13.0, −2.0)	0.015
PASP (mmHg)	35.0 (30.0, 44.5)	41.0 (35.0, 48.0)	48.0 (34.0, 54.0)	0.003
Δ PASP	−7.5 (3.0, 14.5)	−14.5 (8.5, 18.0)	−11.0 (7.0, 24.0)	0.009
SvO ₂ (%)	69.5 (63.0, 75.0)	62.0 (56.4, 70.0)	59.0 (48.0, 65.0)	<0.001
CI (l/min/m ²)	2.8 (2.4, 3.4)	2.5 (2.0, 2.8)	2.2 (1.8, 2.4)	<0.001
Δ CI (l/min/m ²)	+0.3 (−0.1, 0.6)	+0.3 (0.0, 0.6)	+0.4 (0.2, 0.6)	0.26
PVR	1.3 (1.0, 1.7)	1.8 (1.5, 2.2)	2.5 (2.0, 3.3)	<0.001
Δ PVR (WU)	−0.4 (0.2, 0.6)	−1.3 (0.5, 1.5)	−1.8 (1.1, 3.5)	< 0.001
PVR >2 WU	5 (11%)	18 (41%)	33 (77%)	< 0.001
PVR >5 WU	0 (0%)	0 (0%)	1 (2%)	0.36
Δ PVR (%)	−20 (10, 40)	−40 (20, 50)	−50 (30, 70)	< 0.001
Δ PVR >20%	26 (59%)	31 (70%)	36 (84%)	0.040
PA Compliance (mL/mmHg)	3.6 (2.8, 4.8)	2.3 (2.1, 3.2)	1.7 (1.4, 2.2)	<0.001
Δ PA Compliance	+1.1 (0.5, 2.1)	+1.0 (0.5, 1.7)	+0.9 (0.6, 1.2)	0.41
Δ PA Compliance (%)	40 (20, 100)	80 (40, 140)	90 (50, 150)	<0.001
PA Compliance (<2 mL/mmHg)	4 (9%)	6 (14%)	31 (72%)	<0.001
PA Elastance (mmHg/mL)	0.4 (0.3, 0.6)	0.6 (0.5, 0.7)	0.8 (0.6, 1.1)	<0.001
Δ PA Elastance (mmHg/mL)	0.2 (0.1, 0.3)	0.4 (0.2, 0.7)	0.7 (0.5, 1.0)	<0.001
Δ PA Elastance (%)	−25.0 (17.9, 35.9)	−41.9 (19.1, 56.3)	−45.8 (31.4, 56.5)	<0.001
Outcomes				
Hospital LOS (days)	5.1 (4.0, 8.0)	5.5 (4.0, 9.1)	6.8 (5.0, 14.0)	0.15
ICU LOS (days)	2.7 (2.0, 3.2)	3.0 (2.0, 3.8)	3.0 (2.0, 4.0)	0.72
Hemodynamic Deterioration	2 (5%)	1 (2%)	3 (7%)	0.59
Resuscitated cardiac arrest	0 (0%)	0 (0%)	0 (0%)	
90-day Readmission	3 (7%)	1 (2%)	6 (14%)	0.14
PE Death (≤30 days)	2 (5%)	0 (0%)	0 (0%)	0.14
Composite Outcome	4 (9%)	3 (7%)	9 (21%)	0.098

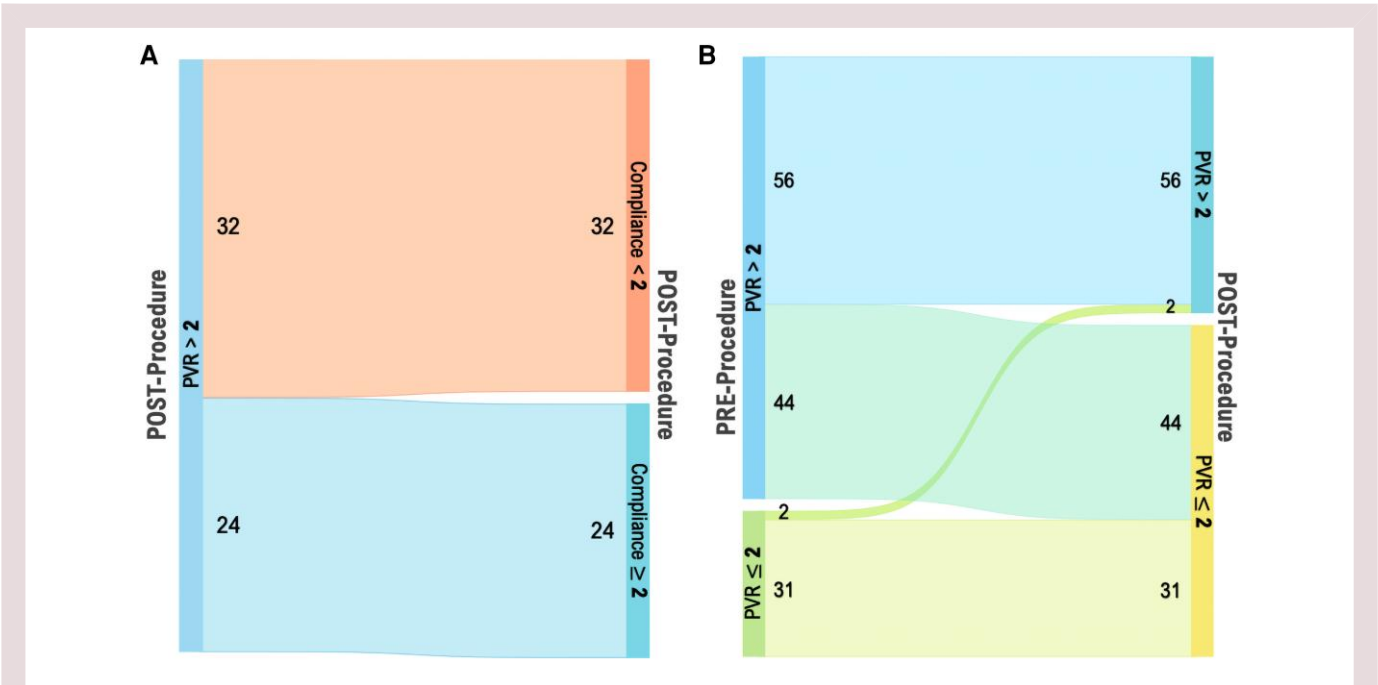
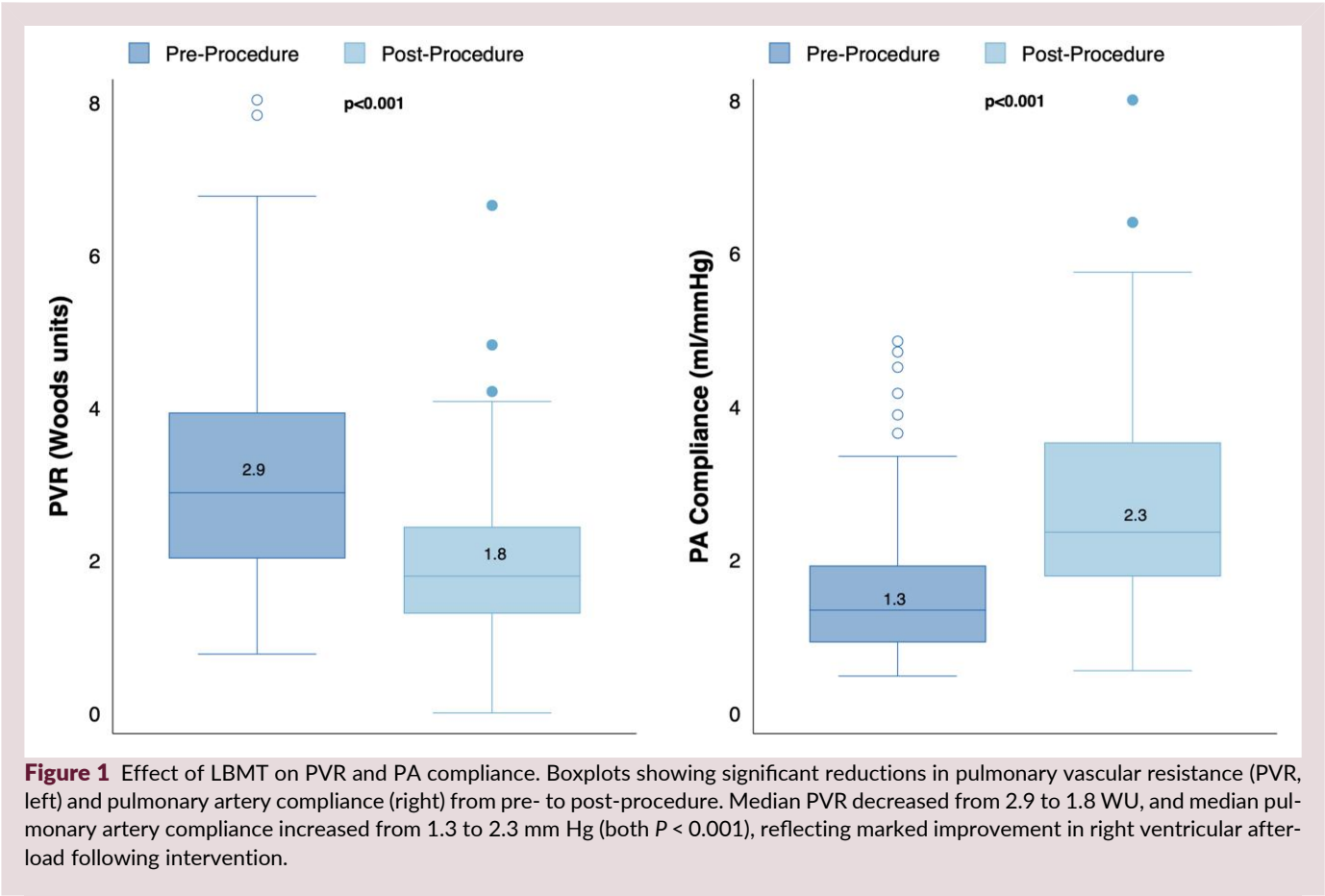
All values are presented as frequency (%) or median (IQR). Bolded values denote statistically significant p-values.

Tertile 1: 0.77–2.28 Wood units (WU), Tertile 2: 2.29–3.71 WU, Tertile 3: 3.72–8.03 WU.

CI, Fick Cardiac Index; ICU, Intensive Care Unit; LOS, length of stay; mPAP, Mean Pulmonary Artery Pressure; Mixed venous oxygen saturation: SvO₂; PA, Pulmonary artery; PASP, Pulmonary Artery Systolic Pressure; PE, Pulmonary Embolism; PVR, Pulmonary Vascular Resistance; WU, Wood Unit.

pulmonary hypertension,²⁴ but the threshold of residual obstruction that warrants further intervention is not well defined. Additionally, procedural risks such as distal embolization, cardiac or pulmonary artery injury, and overall length of procedure must be balanced against the potential benefits of further clot extraction.

Our results highlight the potential mechanism of haemodynamic benefit of LBMT. LBMT significantly reduced PVR with an even greater improvement in PA compliance when compared with baseline. Although 44% of patients had persistently elevated PVR (>2 WU) post-procedure, this measurement was



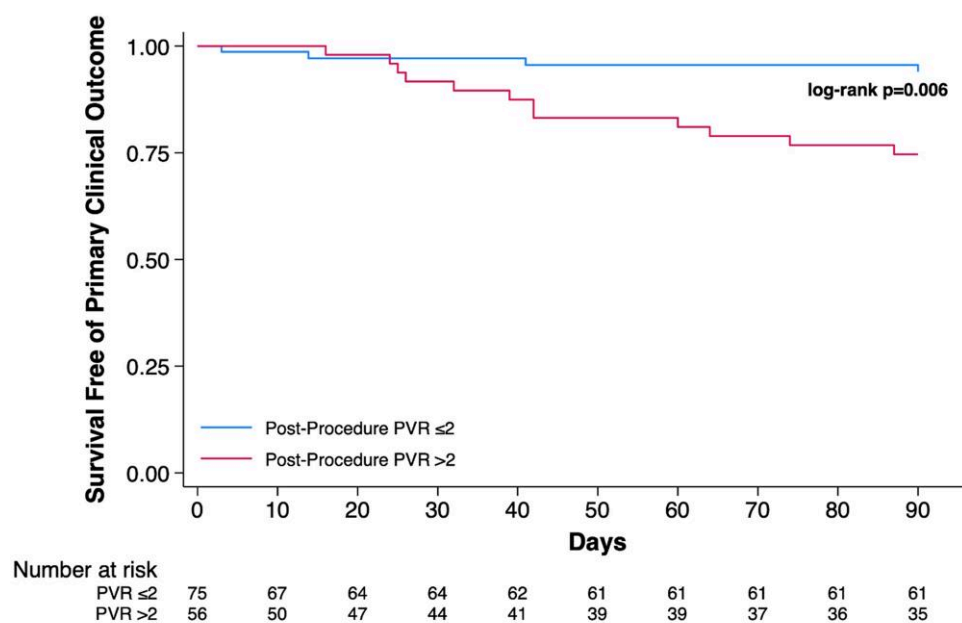


Figure 3 Association of elevated post-procedural PVR with the risk of primary outcome. Kaplan-Meier curve demonstrating significantly lower survival free of the primary outcome in patients with post-procedural PVR >2 WU compared to those with PVR ≤2 WU.

obtained immediately post-thrombectomy and it is likely that a good proportion of these patients would continue to have improvement in PVR several hours to days post-procedure while on anticoagulation. Congruent with this, despite an elevated PVR, 43% of patients had normal PA compliance, suggesting faster recovery of PA compliance than PVR. Overall, 26% of the cohort had persistently elevated PVR >2 WU and low PA compliance immediately post-procedure. It remains to be explored whether this cohort will have improvements in both of these parameters, hours to days post-procedure and on anticoagulation alone. The underlying mechanisms of residual elevated PVR in the immediate post-procedural period are likely multifactorial. While residual obstructive clot burden certainly contributes, acute vasoconstriction and microthrombi, mediated by hypoxia and inflammatory mediators, may also play a substantial role. LBMT primarily addresses proximal mechanical obstruction, whereas distal arteriolar vasoconstriction and microthrombi may recover more slowly, potentially explaining why some patients demonstrate persistently elevated PVR yet near-normal PA compliance. Finally, the highest tertile of baseline PVR had the highest proportion of patients with residual elevated PVR >2 WU, but yet had 0% in-hospital PE-related mortality, perhaps attesting to continued haemodynamic improvement post procedure. Future studies should evaluate the impact of residual pulmonary haemodynamic abnormality on the long-term incidence of CTEPH in this cohort.

Our results highlight, for the first time, the prognostic significance of residual elevated PVR (>2 WU) following LBMT, suggesting that post-procedural PVR may help identify patients who remain at higher risk despite initial clot extraction. Although lower post-procedural PVR was generally associated with more favourable outcomes, the clinical implications of this observation remain uncertain. Importantly, our data do not establish whether post-procedural residual PVR is a

modifiable treatment target or primarily a marker of underlying disease severity. Moreover, we were unable to account for how operators determined procedural completeness, whether guided by angiographic assessment of clot burden, haemodynamics, or other real-time considerations. As a result, the extent to which PVR could inform decisions regarding procedural cessation or the use of adjunctive therapies (such as combining LBMT with catheter-directed thrombolysis) remains unknown. Accordingly, while post-procedural PVR may have potential utility in post-LBMT risk stratification, its role as a procedural endpoint is unproven and will require prospective validation to determine whether PVR-guided strategies can meaningfully influence clinical outcomes.

We also identified several clinical and imaging predictors of post-procedure PVR >2 WU. After adjusting for pre-procedural haemodynamics, the only predictors of residual elevated PVR were pre-procedural mPAP and PVR.

Finally, our results serve as a clear illustration that intermediate-high risk PE represents a physiologically heterogeneous group. Despite similar clinical classification by current ESC guidelines, these patients had markedly different haemodynamic profiles, underscoring the limitations of current risk stratification strategies and the importance of the clinical expertise of PERT.²⁵

Limitations

These findings must be interpreted in the context of several limitations. First, this was a single-centre, retrospective study of patients referred for LBMT, and was subject to selection bias. Procedural strategy, including the duration and extent of thrombectomy, was operator-dependent and not standardized. Second, the use of PADP as a surrogate for PCWP, while previously validated and generally correlates well,¹¹ may introduce measurement variability. However, PADP was used consistently

Table 3 Univariable and multivariable logistic regression for predictors of post-procedural PVR >2 WU

	Odds Ratio	95% CI	P-value	Odds Ratio	95% CI	P-value
Demographics						
Age	1.02	0.99–1.04	0.14	1.02	0.98–1.06	0.21
Male	1.01	0.51–2.01	0.97			
BMI (kg/m ²)	0.99	0.95–1.04	0.66			
Race	1.04	0.76–1.42	0.80			
Hypertension	0.54	0.27–1.08	0.07			
Chronic Lung Disease	3.00	1.05–8.56	0.03	3.25	0.78–13.43	0.10
CKD (III–V)	0.54	0.16–1.86	0.31			
Malignancy	0.54	0.16–1.83	0.30			
Clinical Presentation						
Heart Rate (bpm)	1.00	0.98–1.02	0.89			
Systolic BP (mm Hg)	0.99	0.97–1.00	0.10			
SpO ₂ (%)	0.98	0.92–1.04	0.49			
Syncope	0.79	0.34–1.84	0.57			
Lactate (mmol/L)	1.30	1.03–1.65	0.017	1.19	0.91–1.55	0.20
HS-Troponin (ng/L)	1.00	1.00–1.00	0.20			
Pulmonary Embolism Risk Scores						
CPES Score	1.10	0.97–1.23	0.13			
sPESI	2.97	0.63–14.02	0.16			
Echocardiography						
TAPSE (cm)	1.00	0.69–1.45	0.99			
PASP (mm Hg)	1.03	1.00–1.06	0.06			
McConnell Sign	1.26	0.67–2.38	0.46			
LVOT VTI (cm)	0.91	0.81–1.03	0.12			
Computed Tomography (CT)						
PE Location (Main PA/Lobar)	2.68	0.64–11.24	0.16			
RV/LV Diameter Ratio	3.16	1.33–7.47	0.006	1.07	0.35–3.22	0.90
Pre-Thrombectomy						
CI (L/min/m ²)	0.25	0.12–0.50	< 0.001	1.12	0.37–3.34	0.84
mPAP (mm Hg)	1.07	1.03–1.12	< 0.001	1.07	1.01–1.14	0.026
PVR (WU)	2.43	1.70–3.48	< 0.001	2.20	1.20–4.04	0.011

Bolded values denote statistically significant *p*-values.

BMI, Body Mass Index; BNP, B-type Natriuretic Peptide; BP, Blood Pressure; CI, Fick Cardiac Index; CKD, Chronic Kidney Disease; CPES, Composite Pulmonary Embolism Severity; HS, high-sensitivity; LV, Left Ventricle; LVOT VTI, Left Ventricular Outflow Tract Velocity Time Integral; mPAP, Mean Pulmonary Artery Pressure; PA, Pulmonary Artery; PASP, Pulmonary Artery Systolic Pressure; PE, Pulmonary Embolism; PVR, Pulmonary Vascular Resistance; RV, Right Ventricle; sPESI, Simplified Pulmonary Embolism Severity Index; TAPSE, Tricuspid Annular Plane Systolic Excursion; WU, Wood Unit (mmHg × min/L)

for both pre- and post-procedural calculations of PVR, allowing for internally consistent comparisons despite potential limitations in absolute accuracy. Third, in very obese individuals, the Fick method for estimating cardiac output using assumed or estimated oxygen consumption (VO_2) values may be inaccurate. Obesity is associated with altered metabolic rates and body composition, which can significantly affect VO_2 and, in turn, lead to misestimation of cardiac output. However, only 14.5% of our patient was morbidly obese ($\text{BMI} > 40 \text{ kg/m}^2$). Fourth, the number of outcome events in our cohort was small, and thus we were only able to adjust for age and sex in our CPH model to avoid overfitting. Other clinical, imaging, or haemodynamic factors not included

in the model may be relevant predictors of adverse outcomes and could attenuate or modify the observed association between post-procedural PVR and risk. Fifth, our sample size was not powered to detect differences in mortality or long-term outcomes (such as CTEPH), and large, prospective validation studies are warranted. Finally, although we observed a clear reduction in PVR following LBMT based on direct pre- and post-procedural measurements, the study did not include a comparative control group. We were therefore unable to evaluate how changes in PVR with LBMT compare with the expected haemodynamic trajectory in patients who were treated with anticoagulation alone or with other catheter-based treatment strategies.

Conclusion

In patients with intermediate-risk PE, LBMT was associated with a significant reduction in PVR and improvement in PA compliance immediately post procedure, with an exceedingly uncommon incidence of severely elevated PVR following the procedure. Elevated post-procedural PVR (>2 WU) was independently associated with adverse clinical outcomes; however, its utility as a procedural endpoint remains uncertain and will require prospective studies to determine whether PVR-guided decision-making can meaningfully improve outcomes.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

Author contributions

Robert S. Zhang (Conceptualization, Writing—original draft, Formal analysis, Methodology), Peter Zhang (Data curation, Visualization, Writing—review & editing), Eugene Yuriditsky (Data curation, Writing—review & editing), Lily Jin (Writing—review & editing, Visualization), Felix Mahfoud (Writing—review & editing), Radu Postelnicu (Writing—review & editing), Irene Lang (Writing—review & editing), Carlos L. Alviar (Writing—review & editing), Rachel P. Rosovsky (Writing—review & editing), Daniel Burkoff (Writing—review & editing), and Sripal Bangalore (Conceptualization, Writing—review & editing, Supervision)

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EY, RSZ, PZ, LJ, RP, CLA, report no conflict of interest.

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Data availability

Data supporting the findings are available from the corresponding author upon reasonable request.

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